

[TEGMARK] Falsifiable-predictions / DUNE package + lepton-sector re-tier (my 29g lanes a+b)

Grace | 2026-07-30 | *parallel deep lanes (Casey): (a) the falsifiable-predictions bundle with falsification conditions, headlined by the $0\nu\beta\beta$ color-duality KILL-TEST; (b) a re-tier reconcile of the lepton sector after the K1020 θ_{23} re-ruling. Real curation — a program is only as strong as the list of things that would kill it.*

LANE A — the falsifiable-predictions package (each with its falsification condition)

★ The strong one: $0\nu\beta\beta$ as the color-duality KILL-TEST (pred_004, enhanced)

The color-duality theorem (T2535, Derived this arc) — large-PMNS/small-CKM from color — requires leptons carry a Majorana condensate (a color singlet, which quarks structurally cannot form: $3\otimes 3 = 6\oplus\bar{3}$, no singlet).
- Prediction: $0\nu\beta\beta$ occurs; neutrinos are Majorana. - ★ KILL-TEST: if $0\nu\beta\beta$ is never seen AND neutrinos turn out Dirac, the color-duality explanation is REFUTED. This is a *positive kill-test for a derived theorem* — the strong kind of falsifiability, not a bare null. - Experiment: LEGEND-1000, nEXO, KamLAND-Zen. Timeline: 2026-2035.

θ_{23} octant → DUNE (pred_121, new)

- Prediction: $\sin^2\theta_{23}$ near-maximal ($1/2$, Shilov- $\mathbb{Z}_2$ μ - τ symmetry, Derived), upper-leaning; deviation offset $1/14 = 1/(\text{rank}\cdot g)$ (Identified). Value-form $4/7$ (upper octant).
- Falsification: DUNE resolves the lower octant ($\sin^2\theta_{23} < 0.5$) decisively → μ - τ -breaking direction wrong; OR θ_{23} far from maximal → the Shilov- $\mathbb{Z}_2$ source is wrong.
- Experiment: DUNE, Hyper-K. Timeline: 2027-2035.

δ _PMNS (pred_122, new) — CP frontier

- Prediction: leptonic CP phase; magnitude Derived (banked), branch Structural. The genuine last-frontier lepton observable.
- Falsification: δ _PMNS clearly inconsistent with the banked magnitude → PMNS-phase mechanism wrong. Flagged for the CP-frontier lane.

The Five Absences (each a standing falsifier — any positive detection refutes)

Absence	Falsifier (positive detection refutes)	Corpus
No GUT / proton decay	proton decay observed (τ_p finite)	Five-Absence Set
No dark-matter particle	a WIMP/axion DM particle detected	Five-Absence Set
No magnetic monopoles	a monopole detected	Five-Absence Set
No sterile neutrinos	a sterile- ν confirmed (short-baseline)	Five-Absence Set
No SUSY	a superpartner at any collider	Five-Absence Set

These are the geometry's *irreducibility* predictions (D_IV⁵ has no room for them). Any one positive detection is a clean refutation.

LANE B — lepton-sector re-tier reconcile (post-K1020)

Verified the ledger is consistent after this arc's rulings: | Observable | Current tier | Reconcile note | |—|—|—| θ_{23} | near-maximal DERIVED (Shilov $\mathbb{Z}_2$) + $1/14$ deviation IDENTIFIED | K1020 re-rule applied to $\sin^2\theta_{23}$ _PMNS + T2534; the $4/7$ moment-ratio is a *coincidence*, not the derivation (my K1018 over-claim, corrected) | | θ_{13} | DERIVED ($1/45$) | unchanged, unaffected | | θ_{12} | IDENTIFIED | unchanged | | δ _PMNS | magnitude DERIVED

/ branch STRUCTURAL | frontier (pred_122) | | m_1 (lightest ν) | = 0 (rank-2 M_ν) | unchanged | | color-duality (large-PMNS/small-CKM) | DERIVED (shape) | T2535, this arc's strong result — exact fractions stay Identified; UNTOUCHED by the θ_{23} re-rule | - Consistency: clean. The K1020 θ_{23} re-rule is a single tier correction; θ_{13} , m_1 , the color-duality, and the whole quark sector are unaffected. No orphaned “4/7-is-the-angle” claims remain in the ledger or graph (T2534 corrected).

Net

The lepton sector now ships with a falsifiable-predictions bundle: a headline kill-test ($0\nu\beta\beta$ refutes the color-duality if null+Dirac), a DUNE octant prediction (near-maximal upper-leaning), the CP frontier (δ_{PMNS}), and the Five Absences — each with an explicit falsification condition. Predictions banked to `bst_predictions.json` (pred_004 enhanced, pred_121, pred_122). Re-tier reconcile clean.

— Grace, 2026-07-30 [TEGMARK]. Falsifiable-predictions/DUNE package: $0\nu\beta\beta$ = the color-duality kill-test (null+Dirac refutes T2535 — the strong kind); θ_{23} octant (near-maximal upper-leaning, DUNE); δ_{PMNS} (CP frontier); Five Absences (each a standing falsifier). Banked to `bst_predictions.json`. Lepton-sector re-tier reconcile clean post-K1020 (θ_{23} near-maximal DERIVED + 1/14 IDENTIFIED; color-duality DERIVED untouched).